J-C-118a
August 18, 1965

SUPERSEDING
Int. Fed. Spec. J-C-00118 (Navy-Docks)
September 25, 1964

FEDERAL SPECIFICATION

CABLE, SPECIAL PURPOSE, ELECTRICAL AND WIRE, ELECTRICAL; SHOT FIRING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cable and wire, with a vinyl-polymer or moisture-retardant cotton covering as specified herein, suitable for use in firing explosive charges electrically.

1.2 Classification. Wire and cable covered by this specification shall be one of the following classes, as specified (see 6.1):

Class 1—Single conductor, No. 20 American Wire Gage (AWG), insulated and jacketed with vinyl-polymer compound.
Class 2—Single conductor, No. 14 AWG, insulated and jacketed with vinyl-polymer compound.
Class 3—Two conductor, No. 14 AWG, parallel conductors, insulated and jacketed with vinyl-polymer compound.
Class 4—Single conductor, No. 20 AWG, covered with two cotton servings.
Class 5—Single conductor, No. 14 AWG, covered with two cotton servings and braided jacket.
Class 6—Two conductor, No. 14 AWG, parallel, each conductor covered with two cotton servings and braided jacket and the pair also covered with braided jacket.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specification:

J-C-98—Cable and Wire, Insulated; Methods of Sampling and Testing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 406—Plastics: Methods of Testing.
Fed. Test Method Std. No. 601—Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New

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York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-12000—Cable, Cord, and Wire, Electric, Packaging and Packing of.

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-810—Environmental Test Methods for Aerospace and Ground Equipment.

(Copies of Miliatry Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Standards Association, Inc. (ASA) Standard:

C7.1—Specification for Soft or Annealed Copper Wire. (ASTM B3)

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, N. Y., 10016.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product he sells on the commercial market. Material shall be new and free from all defects and imperfections that may affect the serviceability and appearance of the finished product.

3.2 Conductor. Conductors shall be composed of solid, soft, or annealed copper wire conforming to ASA C7.1. The diameter of No. 14 AWG wire shall be nominal 0.0641 inch and of No. 20 AWG wire shall be nominal 0.0320 inch.

3.3 Lengths. Unless otherwise specified (see 6.1), classes 1 and 4 wire shall be furnished on spools in lengths of 1200 feet, and classes 2 and 5 wire, and 3 and 6 cable in coil lengths of 500 feet.

3.4 Classes.

3.4.1 *Class 1 wire.* Class 1 wire shall consist of a solid copper, No. 20 AWG conductor with vinyl-polymer compound insulation and jacket.

3.4.2 *Class 2 wire.* Class 2 wire shall consist of a solid copper, No. 14 AWG conductor with vinyl-polymer compound insulation and jacket.

3.4.3 *Class 3 cable.* Class 3 cable shall consist of two, class 2 wires laid parallel and enclosed in a vinyl-polymer compound jacket, constructed to permit separation of the individual wires for any given length, when split at the end and intentionally separated.

3.4.4 *Class 4 wire.* Class 4 wire shall consist of a solid copper, No. 20 AWG conductor covered with two helical servings of cotton yarn applied in opposite directions.

3.4.5 *Class 5 wire.* Class 5 wires shall consist of a solid copper, No. 14 AWG conductor, covered with two helical servings of cotton yarn applied in opposite directions and an overall cotton braid, saturated with compounds.

3.4.6 *Class 6 cable.* Class 6 cable shall consist of two class 5 wires laid parallel and enclosed in an overall cotton braid, saturated with compounds.

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3.5 Vinyl-polymer compound insulation and jacket. The insulation and jacket shall be a fungus resistant, vinyl-polymer compound.

3.5.1 *Tensile strength.* The tensile strength of the vinyl-polymer compound shall be not less than 1500 pounds per square inch (p.s.i.), when tested in accordance with 4.4.5.

3.5.2 *Elongation.* The elongation of the vinyl-polymer compound at the breaking point shall be not less than 150 percent, when tested in accordance with 4.4.5.

3.5.3 *Heat resistance.* The vinyl-polymer compound shall not become soft or tacky, nor have a blocking degree greater than slight blocking, when tested at 160°F.

3.5.4 *Dielectric withstand voltage.* The finished vinyl-polymer compound covered wire or cable shall be capable of withstanding a 60 cycle alternating potential of 1500 volts for a period of one minute without evidence of breakdown, when tested in accordance with 4.4.1.

3.5.5 *Insulation resistance.* The minimum acceptance value for insulation resistance for vinyl-polymer compound covered wire or cable shall be 200 megohms per 1000 feet of wire or cable, when a direct current potential of between 200 and 500 volts is applied for one minute at room temperature and tested in accordance with 4.4.2.

3.5.6 *Vinyl-polymer covering flexibility.* The finished vinyl-polymer compound covered wire or cable shall withstand a temperature, unless otherwise specified (see 6.1), of -10°F. without cracking, chipping, or otherwise being damaged.

3.5.7 *Color.* Unless otherwise specified (see 6.1), the color of vinyl-polymer compound shall be yellow.

3.6 Cotton covering. The cotton yarn and braid shall be clean and dry, fungus treated to resist the effects of the test covered in MIL-STD-810, method 508.1, homogeneous, and free from impurities which would adversely affect its physical or electrical properties. The cotton shall be so processed that lumps or irregularities will not appear in the finished covering. The cotton shall be thoroughly dried prior to application of the saturant. The servings shall be closely laid and shall have a circular cross section, concentric about the conductor. The servings and braids shall fit tightly to the conductor or adjacent serving. Braids shall be closely woven. The minimum size and ply of yarn used shall be 14/1 or 30/2. The application of servings shall be such as to give not less than 80 percent coverage of the surface to which applied, when computed as specified in 4.4.1. The application of the braid shall be such that the coverage in each direction shall be not less than 76 percent when computed by means of the following formula:

$$Q = \frac{\text{Net} \times 100}{\text{Sin } A}$$

where Q = Percent coverage in one direction.
N = Number of picks per inch.
e = Number of ends per pick.
t = Diameter of one end of yarn in inches.
A = Lay angle.

3.6.1 *Saturants.* The saturant used in the servings shall be an airblown, black petroleum asphalt, and shall not be mixed with any volatile materials such as paraffin, wax tailings, scale wax, or flux oils. Braids shall be saturated with paraffin. Saturation shall be such that diagonal cuts through the braid or servings will not expose unsaturated fibers.

3.6.2 *Finish.* Class 4 wire shall have a black finish and shall be free of any tackiness. The paraffin saturated braids of class 5 wire and class 6 cable shall have a polished finish and be of the color specified (see 6.1).

3.6.3 *Cotton yarn flexibility.* The yarn of the coverings shall not break when the wire is subjected to the flexibility test de-

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scribed in 4.5.1. Breaking or cracking of the finish or saturant shall not constitute a failure.

3.7 **Marking.** Each spool or coil of wire or cable shall be marked or tagged to indicate the manufacturer's name, date of manufacture, grade, length in feet, and AWG size.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for lot acceptance.** Formation of lots, and sampling for lot acceptance shall be in accordance with J-C-98.

4.3 **Examination.** Each sample selected in accordance with 4.2 shall be examined for compliance with the requirements specified in section 3. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.4 **Product tests.** Selected samples of wire and cable shall be tested as specified in 4.4.1 through 4.4.8. Certified copies of all test results shall be furnished. The Government will elect to witness the tests, or to accept the test certification as evidence of compliance, or both.

4.4.1 *Vinyl-polymer dielectric strength test.* The dielectric test shall be made on full coils, spools, or reels of cable. A 60 cycle alternating potential of 1500 volts shall be applied between conductors for a period of one minute at normal room temperature. The dielectric strength shall meet the requirements of 3.5.4.

4.4.2 *Vinyl-polymer insulation resistance test.* Wires and cables shall be tested for insulation resistance as soon as practicable after completion of the dielectric test. The leakage current shall be measured after a one minute electrification with a direct current potential of not less than 200 nor more than 500 volts applied between conductors and from each conductor to ground while wire or cable is immersed in water. The insulation resistance shall meet the requirements of 3.5.5.

4.4.3 *Vinyl-polymer covering flexibility test.* The specimen shall be a piece of finished wire or cable 7-1/2 ± 1/2 inches long. All samples, together with all testing equipment shall be conditioned for 5 hours at the specified temperature (see 3.5.6). A 3/8 inch mandrel for No. 14 AWG wire or cable or a 1/4 inch mandrel for No. 20 AWG wire shall be securely clamped at both ends in a horizontal position to prevent rotation. The specimen shall be wrapped twice around the mandrel. A 5-kilogram weight shall be suspended from one end of the cable, by a heavy-duty battery clip attached to the weight, and allowed to drop freely, unwinding the wire from the mandrel. The two turns of wire shall be completely in contact with the mandrel prior to releasing the weight so that the wire will be taut when unwinding. The free end shall not be longer than 1/2 inch to permit rapid unwinding of the wire from the mandrel. The mandrel shall be positioned at a height above the base sufficient to permit a wire to drop free after unwinding from the mandrel. The samples and test equipment shall have been maintained at the specified temperature during the test, and the samples shall be left undisturbed at that temperature for 5 minutes thereafter. The samples shall then be examined visually for any cracks. Any cracks found shall be cause for rejection of the lot represented by the samples.

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4.4.4 *Thickness of vinyl-polymer.* Sufficient minimum thickness of vinyl-polymer compound shall be determined by using inspection equipment specified in Fed. Test Method Std. No. 601.

4.4.5 *Tensile strength and elongation.* Tensile strength and elongaton of the vinyl-polymer compound shall be determined on dumbell specimens in accordance with test methods 1012 and 1013 of Fed. Test Method Std. No. 406 from sheets of the same material and treatment used in wire or cable manufacture. The tensile strength shall meet the requirements of 3.5.1 and the elongation 3.5.2.

4.4.6 *Heat resistance.* Resistance to heat of the vinyl-polymer covering shall be determined by a blocking test conducted in the following manner: Specimens shall be tested in accordance with method 1131 of Fed. Test Method Std. No. 406 at a temperature of 160° ± 2°F. (71° ± 1° C.) with an applied pressure of one-third p.s.i. The heating shall be for 48 hours. The required test specimens shall be cut from the sheet used to obtain the tensile strength specimens.

4.4.7 *Cotton covering tests.* Tests shall be conducted in accordance with J-C-98, test methods as indicated in table I, to determine conformance to the covering requirements specified in 3.6.

TABLE I. *Covering test methods*

Determination	Symbol	Test method
Serving coverage	—	8131
Number of picks per inch	N	8011
Number of ends per pick	e	8021
Diameter of one end of yarn in inches	t	8041
Lay angle	A	1611

4.4.8 *Cotton covering flexibility test.* Suitable length specimens of completed class 4 or 5 wire and individual conductors of class 6 cable, and a mandrel of the same diameter as the wire, shall be cooled to a temperature of 0°C. held at that temperature for a period of one hour, then the wire wrapped around the mandrel for six complete turns at a uniform rate of not more than ten turns per minute.

4.5 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the wire and cable shall be inspected to verify comformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the levels of preservation, packaging, and packing used are defined in Fed. Std. No. 102.)

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* The wire and cable shall be packaged in accordance with level A of MIL-C-12000.

5.1.2 *Level C.* The wire and cable shall be packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* The wire and cable shall be packed in accordance with level A of MIL-C-12000.

5.2.2 *Level B.* The wire and cable shall be packed in accordance with level B of MIL-C-12000.

5.2.3 *Level C.* The wire and cable shall be packed to insure arrival at destination in satisfactory condition and carrier acceptance at lowest rates in compliance with rules and regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Class required (see 1.2).

(*c*) Length when other than specified (see 3.3).

(*d*) Flexibility temperature requirement for class 1, 2, or 3 when other than as specified in 3.5.6.

(*e*) When other than yellow color is required for class 1, 2, or 3 (see 3.5.7).

(*f*) Color required for class 5 or 6 (see 3.6.2).

(*g*) Responsibility for inspection, if other than specified (see 4.1).

(*h*) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

(*i*) Special marking, if required (see 5.3).

6.2 **Cross reference.** The cross reference between the classification shown in J-C-00118 (Navy-Docks) and this edition is as follows:

	J-C-00118 (Navy-Docks)	*J-C-118a*
Type I	Class A	Class 4
	Class B	Class 5
	Class C	Class 1
	Class D	Class 2
Type II	Class E	Class 6
	Class F	Class 8

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Wire

Classes 1, 2, 4 *and* 5

Rail:

Wire, covered, insulated, copper.
Carload minimum weight 30,000 pounds.

Motor:

Wire, covered, insulated, copper.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Cable

Classes 3 *and* 6

Rail:

Cable, electric, copper, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Cable, electric, copper, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY INTERESTS:

Custodians:
Army—EL
Navy—YD
Air Force—11

Review:
Army—EL
Navy—YD
Air Force—85

User:
Army—MI, MU
Navy—MC
Air Force—17

Preparing activity:
Navy—YD

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INTERESTED CIVIL AGENCIES:

Department of Commerce
General Services Administration
Department of Justice
Post Office Department

Copies of this specification may be purchased for 5 cents each.

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